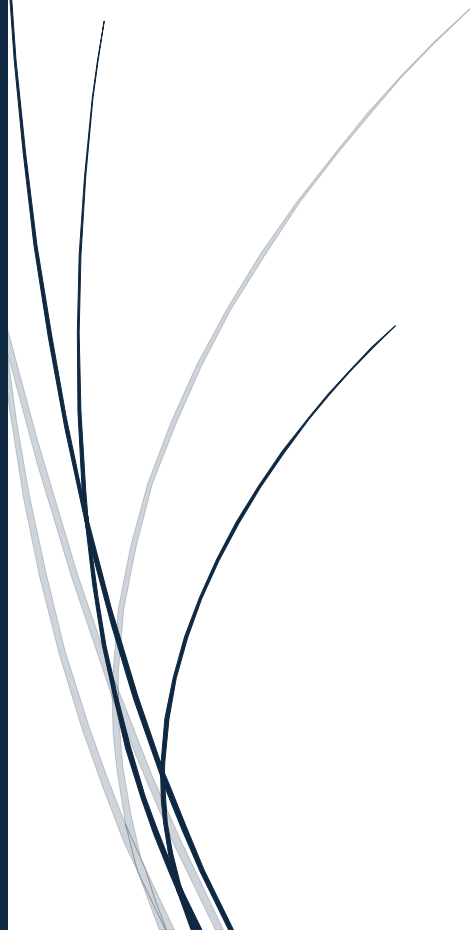




CMPG325

MILESTONE 1

Client Design Review



LB Madikwe
33167044

1. Client Requirements

1.1 Client Information

Item	Details
Organisation	Dijo Tsa Rona Foods Processing (Vryburg)
Client ID	CLI-032
Industry	Manufacturing
Project ID	CMPG325-2026-032
Addressing block	192.168.25.0/24
Main Challenge	IPv4 VLSM subnetting
Wireless requirement	Wireless only in public areas
Change request	CR15 – second internet connection

1.2 Network requirements

The client needs a network that can support the different areas of the food processing organisation.

The network must:

- Connect the office users.
- Connect the production area.
- Provide access to servers.
- Provide a management network for network devices.
- Provide wireless access in the allowed public areas.
- Allow communication between the required networks.
- Use the assigned 192.168.25.0/24 address block.
- Use VLSM to avoid wasting IP addresses.
- Have two internet connections for better reliability.
- Be designed and tested in Cisco Packet Tracer.
- Allow the network to continue using the second internet connection if the main internet connection fails.

1.3 Wireless requirement

Wireless will not be provided across the whole factory.

It will only be available in the specified public areas. This helps to keep the wireless network limited to areas where it is required.

For the design, I have planned a separate Public Wi-Fi network.

1.4 Change request – CR15

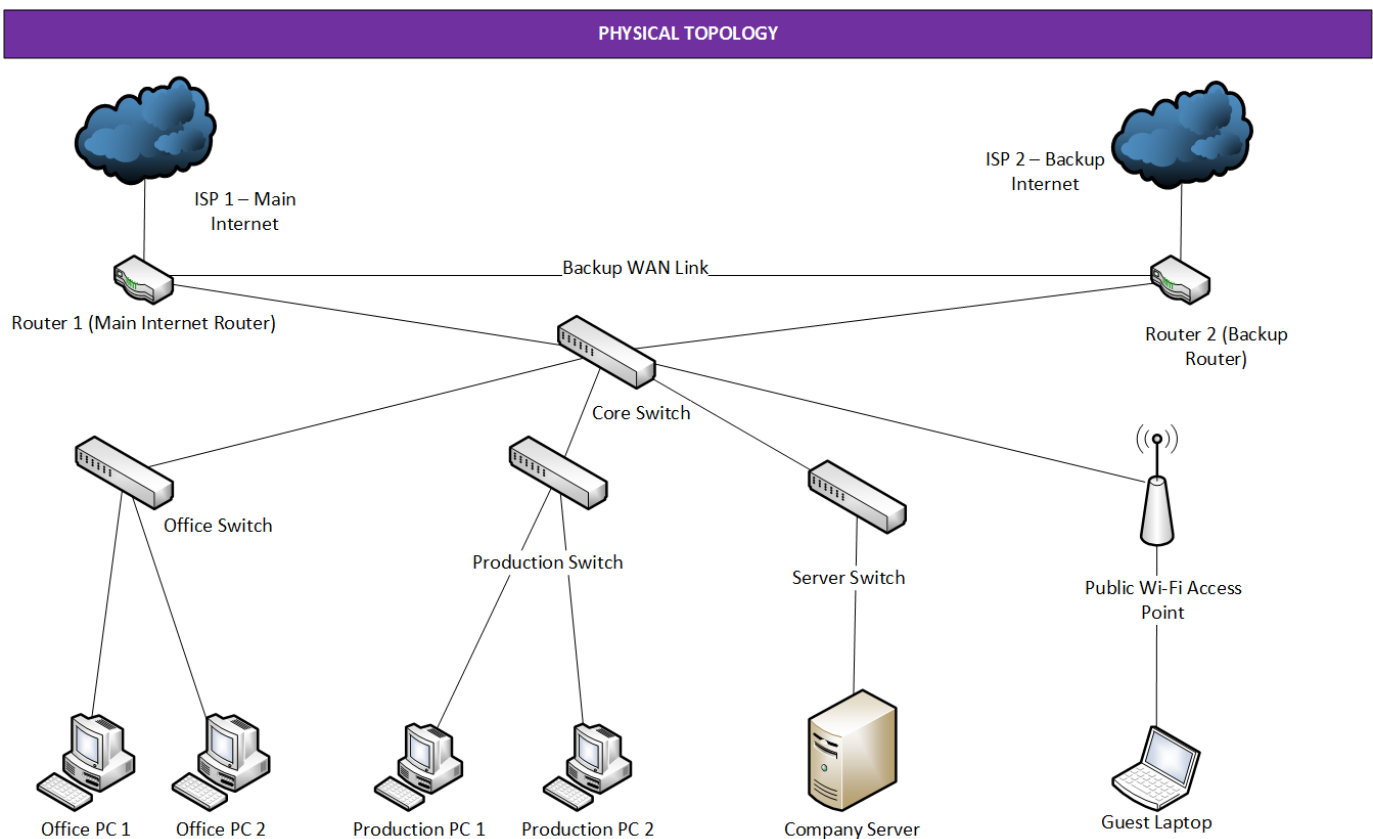
The client requested a second internet connection.

The reason for this is resilience.

The network will have:

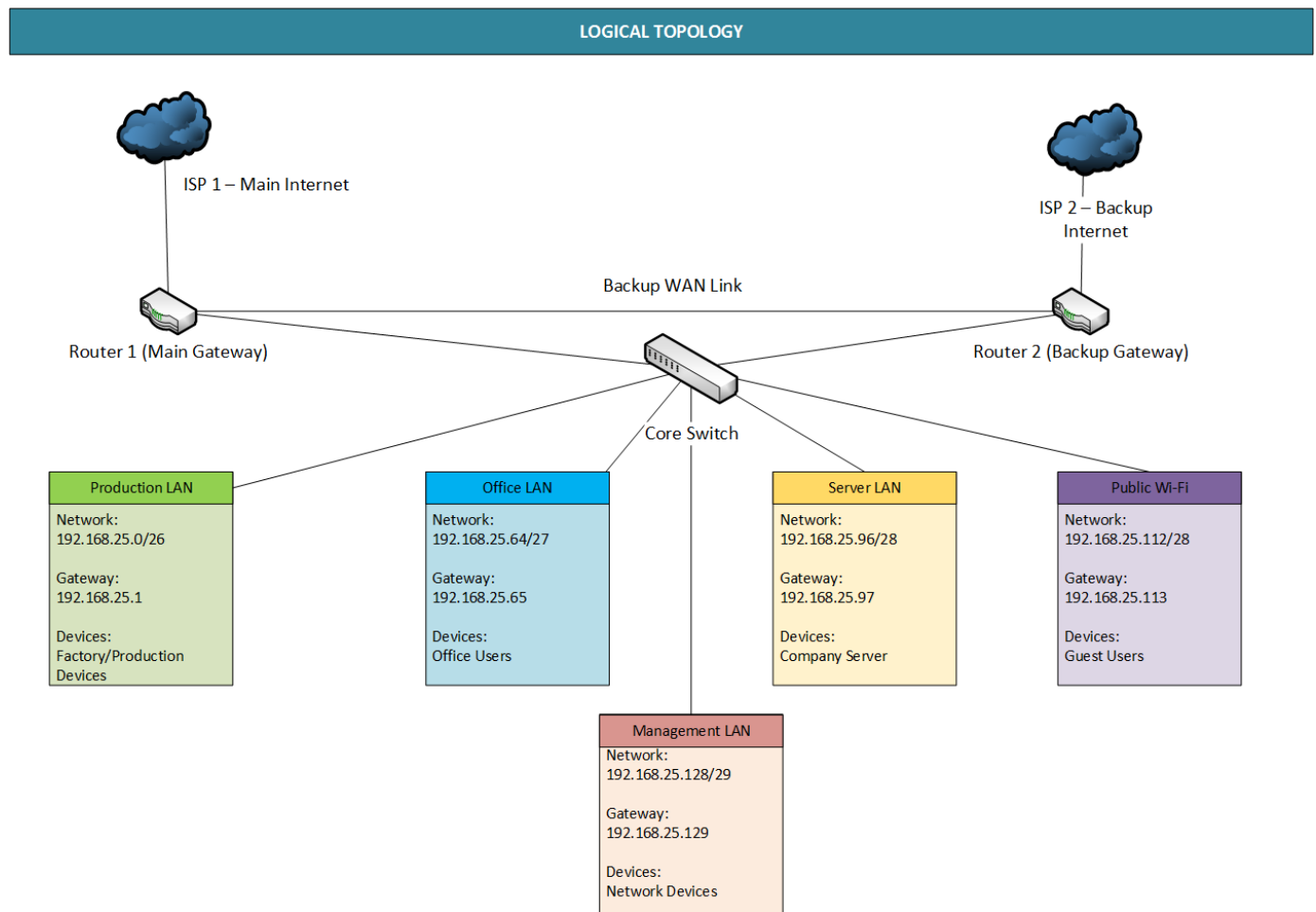
- ISP 1 – main internet connection
- ISP 2 – backup internet connection
- The second connection will be available if the first internet connection has a problem.

2. Physical Topology



The physical topology shows the connection of the network devices. The core switch connects the different departments, while the two internet connections provide reliability through backup connection.

3. Logical Topology



The logical topology separates the network into different VLANs using VLSM addressing. Each department has its own subnet to improve organisation and network management.

4. IP Addressing Plan

Network/ Department	Network Address	Pref ix	Subnet Mask	Usable IP Range	Default Gateway
Production Network	192.168.25.0	/26	255.255.255.192	192.168.25.1 - 192.168.25.62	192.168.25.1
Office Network	192.168.25.64	/27	255.255.255.224	192.168.25.65 - 192.168.25.94	192.168.25.65
Server Network	192.168.25.96	/28	255.255.255.240	192.168.25.97 - 192.168.25.110	192.168.25.97
Public Wi-Fi Network	192.168.25.112	/28	255.255.255.240	192.168.25.113 - 192.168.25.126	192.168.25.113
Management Network	192.168.25.128	/29	255.255.255.248	192.168.25.129 - 192.168.25.134	192.168.25.129
Router-to- Router Link	192.168.25.136	/30	255.255.255.252	192.168.25.137 - 192.168.25.138	N/A

5. Initial Github Repository

Github Repository Link: <https://github.com/lizziebusisiwe0410/CMPG325-Network-Design-Project>